Name: _____

Class: _____

Date: _____ **Mr. Rawson • GCSE Physics**

Hovercraft Demo: Inertia

Two ideas to use

Newton's first law says that an object stays at rest, or keeps moving at the same speed in a straight line, unless a resultant force acts on it.

Inertia is the tendency of an object to keep its motion the same. The more mass an object has, the more inertia it has.



Image: PASCO scientific, Hovercraft ME-9838 manual

The rider sits on the platform and the fan blows air under it. With the fan on, the cushion of air lifts the hovercraft off the floor, so almost no friction acts on it. The two arrows are the same length: the air force up and the weight down are **equal**, so they cancel, and the **resultant force is zero** while it glides.

The same three steps for every demo: **predict** before it happens, **observe** it, then **explain** with the words *resultant force* and *inertia*.

Demo 1: At rest

The hovercraft sits on the floor with the fan on, and nobody touches it. What does it do?

Predict. _____

Observe. _____

Explain. The resultant force on it is _____, so it _____

_____

Demo 2: One push, then let go

Give the hovercraft one push, then let go. Watch its speed and direction after your hand leaves it.

Predict. _____

Observe. _____

Explain. After the push, almost no force acts along the floor, so it _____

_____

Demo 3: Fan off while it moves

Push the hovercraft, then switch the fan off while it glides. What happens, and how fast?

Predict. _____

Observe. _____

Explain. With the fan off, friction acts on it, and the resultant force is _____

_____

Demo 4: Add mass

Load the hovercraft with extra mass, as your teacher directs. Compare how hard it is to start and to stop.

Predict. _____

Observe. _____

Explain. The loaded hovercraft has more mass, so it has more _____

_____